HACKATHON BLUEPRINT

MANGA MY LIFE

2-Hour Implementation & Architecture Guide

1. TECHNOLOGY STACK & DECISION RATIONALE

Frontend Core

React + Vite: Instant setup, lightning-fast HMR, zero-config bundling. Crucial for rapid 2-hour iteration.

Styling & Manga FX

Tailwind CSS: Utility-first styling for speedlines, halftone background patterns, speech bubble tails, and bold borders.

AI Logic Engine

OpenAI / Groq API: Direct client-side SDK fetch using standard structured JSON mode (or system prompt format enforcer).

Export & Sharing

html2canvas + Print API: Single-click DOM rendering to downloadable PNG image with standard window.print() CSS fallback.

2. TACTICAL 2-HOUR EXECUTION PLAN

Timeframe	Task Focus	Deliverable / Milestone
00–15 min	Scaffolding & UI Framework	Vite app live with text input, genre selector, and submit button.
15–35 min	Manga Panel Layout	3-panel vertical CSS grid with speech bubbles, captions, and SFX layers.
35–60 min	AI Engine Integration	LLM prompt connection returning validated structured JSON response.
60–80 min	Visual Polish & Manga Styling	Halftone CSS textures, action lines, dramatic fonts, screen shakes.
80–95 min	Fallback & Example Prompts	Preset buttons + mock JSON fallback ensuring guaranteed demo success.
95–110 min	Export Functionality	html2canvas script integration for downloading final comic strip.
110–120 min	Demo Dry Run	Scripted run with "Villain Mode" reveal rehearsed for judges.

Golden Hackathon Rule: Do not block the pitch on real-time image generation models (DALL-E / Midjourney). They are slow (10s+) and unreliable. Use high-impact CSS assets, background templates, and dynamic typography for instant 2-second renders.

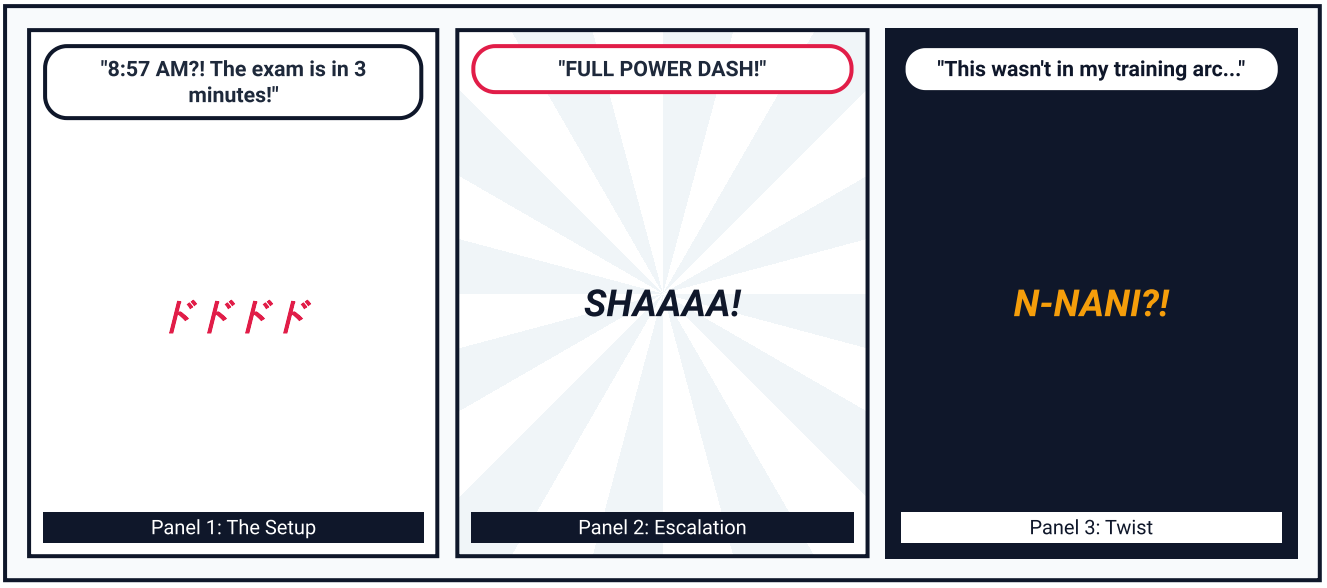
3. STRUCTURED AI SCRIPTWRITER PROMPT

Use direct JSON response enforcement in your LLM call to get predictable data:

```
const SYSTEM_PROMPT = `You are a dramatic manga scriptwriter. Transform the user's life story into a 3-panel manga comic. Genre: {{genre}} | Story: {{story}} Return ONLY valid JSON matching this schema: { "title": "Dramatic Comic Title", "genre": "Selected Genre", "panels": [ { "panel_num": 1, "scene": "Short action summary", "caption": "Narrator note or timestamp", "dialogue": "Character quote (max 12 words)", "sound_effect": "Japanese/English SFX (e.g. ドドド, SHAAAA)", "expression": "shocked | happy | angry | determined", "background": "action_lines | classroom | dramatic_darkness" } ] }`;
```

4. COMPONENT VISUAL ARCHITECTURE

CSS-driven panel structure avoiding external visual dependencies during live demos:



5. THE WINNING DEMO PITCH SCRIPT

- **Hook (0-15s):** "Everyday life feels mundane, but in our minds, missing a bus is an apocalyptic disaster."
- **Live Action (15-45s):** Enter real scenario → Pick *Shonen Mode* → Hit Generate → Show dramatic reveal with sound effects.
- **The Twist Feature (45-75s):** Click **"Villain Mode"** on the same story → Watch the laptop point-of-view transform the story instantly.
- **Closing (75-90s):** "Every life has a story. We just give yours the manga treatment."